

The Weinberg Angle from Hopf Bundle Geometry

A Hopf-Geometric Route to the Same Estimate — and the Same Gap

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

This paper develops the Weinberg-angle estimate through the Hopf bundle structure that the rope two-strand state space carries, offering a second, more geometric route to $\sin^2\theta_W = 1/(3\sqrt{2})$. The Hopf framing makes the origin of the '3' and ' $\sqrt{2}$ ' more transparent, but it reaches the same numerical value and therefore inherits the same $\sim 1.94\%$ gap from measurement. It is reported with the identical honest caveat: suggestive geometry, not a precision derivation. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants.

Scope of this paper

CLAIM: a Hopf-bundle route to $\sin^2\theta_W = 1/(3\sqrt{2})$, clarifying the geometric origin of the factors.

NOT CLAIMED: improved precision. Same $\sim 2\%$ gap as the winding route. Status: DEMOTED 5 September 2026 (KOIDE-WEINBERG, EW-001) to a KEPT COINCIDENCE; the Hopf route clarifies the factors and supplies no forcing mechanism. See the Disposition section.

1. The Hopf Bundle of the Two-Strand State

The rope two-strand state space carries the Hopf bundle structure $S^3 \rightarrow S^2$. Reading the mixing angle from the base/fibre geometry of this bundle reproduces the winding-angle estimate, with the 3 arising from the spatial embedding and the $\sqrt{2}$ from the two-strand fibre.

2. Same Value, Same Gap

The Hopf route yields $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$, identical to the winding-geometry estimate and hence 1.94% from the measured value. The geometric clarity is the contribution; the numerical status is unchanged.

The honest status of this value. The predicted $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ differs from the measured 0.23122 by about 1.94%. It is close, and its form is suggestive, but it is NOT exact, and the test suite pins it explicitly as a 'soft external input' — the value that, when fed into the lepton-

mass relation instead of the measured angle, breaks it. This is therefore a geometric conjecture of the right magnitude, not a precision derivation, and the $\sim 2\%$ gap is the known weak link of the electroweak sector.

Disposition (5 September 2026): demoted to a kept coincidence

Verdict. Commission KOIDE-WEINBERG (ITEM 6d) executed the derivation-or-demotion this paper's status line deferred. Under FND-088's protocol (require a dependency path from registered objects to the number, with every choice point listed, before looking at the target) the chain 'three dimensions supply the 3, two strands supply the $\sqrt{2}$ ' has three choice points on its face: (1) the combination rule, why $1/(3\sqrt{2})$ rather than $1/\sqrt{6}$, $1/3^2$, $\sqrt{2}/3$ or $1/(3+\sqrt{2})$, which the two-axioms paper itself records as 'uniqueness is Open'; (2) the quantity, why $\sin^2\theta_W$ carries the geometric number rather than $\sin\theta_W$, $\tan^2\theta_W$ or g'/g ; (3) the multiplicative reading of the two factors, for which the Hopf route supplies transparency but no forcing mechanism. A chain with choice points is a conjecture whose choices are now named. The registered verdict is EW-DEMOTE: EW-001 leaves Conjecture grade and is kept as a coincidence. Full record: analysis/KOIDE_WEINBERG_results.md.

The renormalization point, fixed before re-reading. A scale-free geometric number can be compared to exactly one renormalization point. The registered comparison, 0.23122 (the MS-bar value at M_Z), was locked before the comparison was re-read; against it the estimate is 1.94% off. The low- Q^2 effective angle (about 0.238) was displayed alongside and never adjudicated against: the estimate is 0.97% from it. A number within one to two percent of whichever point one picks is what a coincidence at that level looks like, and running or threshold effects are not invoked to close the gap.

What stands. The two structural readings (three dimensions, two strands) stand as readings; the 1.94% gap stands as measured; the test-suite pin stands. What is withdrawn is the word 'estimate' as a claim about the electroweak sector: the corpus holds no derived Weinberg angle. The companion lepton-mass relation (PM-001) was demoted in the same commission on independent grounds (its sensitivity to this angle is 71: a 1.94% change in the input moves the muon-to-electron mass ratio by a factor 7.8), which closes the coupled inconsistency the corpus carried between the two entries.

Status

Hopf-geometric $\sin^2\theta_W = 1/(3\sqrt{2})$: DEMOTED to a kept coincidence with the winding route (EW-001, 5 September 2026).

$\sim 1.94\%$ deviation — unchanged; the sector's weak link.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.